# AI-Powered CRM Platform - System Analysis Document

**CNC111 - Web Programming Project**

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## System Overview

The **AI-Powered CRM Platform** is a modern Customer Relationship Management system that leverages artificial intelligence to enhance lead management, appointment scheduling, and customer interactions. The system is built using Next.js 14+, Supabase, and integrates with OpenAI GPT-4o for intelligent automation.

### Key Modules:

- **LeadCatch Module** - AI-powered lead capture, parsing, and enrichment
- **Appointments Module** - Smart scheduling with calendar integrations
- **CRM Core** - Contact management, sales pipeline, and analytics
- **AI Module** - Shared AI services for all modules

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## Use Case Diagram

```
+-------------------+     +-------------------+     +-------------------+
|   User/Sales Rep  |     |       Admin       |     |     AI System     |
+---------+---------+     +---------+---------+     +---------+---------+
          |                         |                         |
          |                         |                         |
+---------|-------------------------|-------------------------|-------------------+
|         v                         v                         v                   |
|  +===========================================================+                  |
|  |                    AUTHENTICATION                          |                 |
|  |  +------------+  +------------+  +---------------+         |                 |
|  |  |   Login    |  |  Register  |  | Reset Password|         |                 |
|  |  +------------+  +------------+  +---------------+         |                 |
|  +===========================================================+                  |
|                                                                                  |
|  +===========================================================+                  |
|  |                   LEADCATCH MODULE                         |                 |
|  |  +---------------+  +------------------+  +-------------+  |                 |
|  |  | Capture Lead  |  | Parse LinkedIn   |  | Import CSV  |  |                 |
|  |  +---------------+  +------------------+  +-------------+  |                 |
|  |  +---------------+  +------------------+  +-------------+  |                 |
|  |  | Enrich Lead   |  |   Score Lead     |  | Export Leads|  |                 |
|  |  +---------------+  +------------------+  +-------------+  |                 |
|  |  +-------------------+  +------------------+                |                |
|  |  | Detect Duplicates |  | Categorize Lead  |                |                |
|  |  +-------------------+  +------------------+                |                |
|  +===========================================================+                  |
|                                                                                  |
|  +===========================================================+                  |
|  |                  APPOINTMENTS MODULE                       |                 |
|  |  +--------------------+  +--------------------+            |                 |
|  |  | Schedule Appointment|  | Manage Availability |           |                |
|  |  +--------------------+  +--------------------+            |                 |
|  |  +---------------+  +---------------+  +----------------+  |                 |
|  |  | Sync Calendar |  | Book Meeting  |  |   Reschedule   |  |                 |
|  |  +---------------+  +---------------+  +----------------+  |                 |
|  |  +---------------+  +------------------+                   |                 |
|  |  |    Cancel     |  |  Send Reminders  |                   |                 |
|  |  +---------------+  +------------------+                   |                 |
|  +===========================================================+                  |
|                                                                                  |
|  +===========================================================+                  |
|  |                      CRM CORE                              |                 |
|  |  +-----------------+  +------------------+  +------------+ |                 |
|  |  | Manage Contacts |  | Manage Companies |  | Manage Deals| |                 |
|  |  +-----------------+  +------------------+  +------------+ |                 |
|  |  +--------------+  +---------------+  +-----------------+ |                 |
|  |  | Create Tasks |  | View Pipeline |  | View Analytics  | |                 |
|  |  +--------------+  +---------------+  +-----------------+ |                 |
|  +===========================================================+                  |
|                                                                                  |
|  +===========================================================+                  |
|  |                    ADMINISTRATION                          |                 |
|  |  +--------------+  +-------------------+  +--------------+ |                 |
|  |  | Manage Users |  | Configure Settings|  | View Reports | |                 |
|  |  +--------------+  +-------------------+  +--------------+ |                 |
|  +===========================================================+                  |
|                                                                                  |
|                         AI-POWERED CRM SYSTEM                                    |
+---------------------------------------------------------------------------------+
          |                         |                         |
          v                         v                         v
+-------------------+     +-------------------+     +-------------------+
| Calendar Service  |     |Enrichment Service|     |  Email Service    |
| (Google/Outlook)  |     | (Apollo/Clearbit)|     | (Resend/Twilio)   |
+-------------------+     +-------------------+     +-------------------+
```

### Use Case Relationships

```
+------------------+         <<includes>>          +------------------+
|  Capture Lead    |------------------------------->|   Enrich Lead    |
+------------------+                               +------------------+
        |                                                   |
        | <<includes>>                                      | <<includes>>
        v                                                   v
+------------------+                               +------------------+
|   Score Lead     |                               | Categorize Lead  |
+------------------+                               +------------------+

+------------------+         <<extends>>           +------------------+
| Parse LinkedIn   |------------------------------->|  Capture Lead    |
+------------------+                               +------------------+

+------------------+         <<extends>>           +------------------+
|   Import CSV     |------------------------------->|  Capture Lead    |
+------------------+                               +------------------+

+------------------+         <<includes>>          +------------------+
|  Book Meeting    |------------------------------->|  Send Reminders  |
+------------------+                               +------------------+
        |
        | <<includes>>
        v
+------------------+
|  Sync Calendar   |
+------------------+
```

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## Narrative Use Cases

### Use Case 1: Capture and Enrich Lead

| Field | Description |
|---|---|
| **Use Case ID** | UC-001 |
| **Use Case Name** | Capture and Enrich Lead |
| **Actor(s)** | User (Sales Rep), AI System, Enrichment Service |
| **Description** | User captures a lead from various sources, and the system automatically enriches it with additional data using AI |
| **Preconditions** | User is logged into the system |
| **Postconditions** | Lead is stored in database with enriched data and AI-generated score |

**Main Flow:**

| Step | Action |
| --- | --- |
| 1 | User accesses the LeadCatch module |
| 2 | User selects input method (LinkedIn URL, Website, Manual, or CSV) |
| 3 | User provides lead information |
| 4 | System validates input data |
| 5 | AI parses and extracts lead information |
| 6 | System initiates waterfall enrichment (Apollo → Clearbit) |
| 7 | AI scores the lead (1-100) |
| 8 | AI categorizes and tags the lead |
| 9 | System checks for duplicates |
| 10 | System saves the lead to database |
| 11 | System displays enriched lead profile to user |

**Alternative Flows:**

| Step | Condition | Action |
| --- | --- | --- |
| 4a | Invalid input | System displays error, returns to step 3 |
| 6a | Service unavailable | System saves partial data, flags for retry |
| 9a | Duplicate found | System prompts user to merge or create new |

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## Use Case 2: Schedule Appointment

| Field | Description |
| --- | --- |
| **Use Case ID** | UC-002 |
| **Use Case Name** | Schedule Appointment |
| **Actor(s)** | User, External Attendee, Calendar Service, AI System |
| **Description** | User or external attendee schedules a meeting through the system |
| **Preconditions** | User has configured availability and event types |
| **Postconditions** | Appointment is created and synced to all calendars |

**Main Flow:**

| Step | Action |
| --- | --- |
| 1 | Attendee accesses public booking page |
| 2 | Attendee selects event type |
| 3 | System retrieves user’s availability from connected calendars |
| 4 | System calculates available time slots |
| 5 | Attendee selects preferred date and time |
| 6 | Attendee enters contact information |
| 7 | AI suggests optimal meeting time (optional) |
| 8 | System validates availability (no conflicts) |
| 9 | System creates appointment in database |
| 10 | System syncs to Google Calendar/Outlook |
| 11 | System sends confirmation email to both parties |
| 12 | System schedules reminder notifications |

**Alternative Flows:**

| Step | Condition | Action |
| --- | --- | --- |
| 5a | No available slots | System suggests alternative dates |
| 8a | Conflict detected | Returns to step 5 with updated availability |
| 10a | Calendar sync fails | System queues for retry, notifies user |

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## Use Case 3: Manage Sales Pipeline

| Field | Description |
| --- | --- |
| **Use Case ID** | UC-003 |
| **Use Case Name** | Manage Sales Pipeline |
| **Actor(s)** | User (Sales Rep), Admin |
| **Description** | User manages deals through different stages of sales pipeline |
| **Preconditions** | User is logged in, deals exist in the system |
| **Postconditions** | Deal status is updated, activities are logged |

**Main Flow:**

| Step | Action |
|---|---|
| 1 | User accesses Pipeline view (Kanban board) |
| 2 | System displays deals organized by stage |
| 3 | User drags deal card to new stage |
| 4 | System updates deal status |
| 5 | System logs activity in timeline |
| 6 | System triggers automated actions (notifications, tasks) |
| 7 | System updates analytics |

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### Use Case 4: Import Leads from CSV

| Field | Description |
|---|---|
| **Use Case ID** | UC-004 |
| **Use Case Name** | Import Leads from CSV |
| **Actor(s)** | User, AI System |
| **Description** | User bulk imports leads from a CSV file |
| **Preconditions** | User has a CSV file with lead data |
| **Postconditions** | All valid leads are imported and enriched |

**Main Flow:**

| Step | Action |
| --- | --- |
| 1 | User accesses Import function in LeadCatch |
| 2 | User uploads CSV file |
| 3 | System validates file format |
| 4 | System displays field mapping interface |
| 5 | User maps CSV columns to lead fields |
| 6 | System validates data in each row |
| 7 | System detects duplicates |
| 8 | User confirms import (with duplicate handling options) |
| 9 | System imports leads in batches |
| 10 | System queues leads for AI enrichment |
| 11 | System displays import summary report |

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### Use Case 5: Generate AI-Powered Email

| Field | Description |
| --- | --- |
| **Use Case ID** | UC-005 |
| **Use Case Name** | Generate AI-Powered Email |
| **Actor(s)** | User, AI System |
| **Description** | User generates personalized email content using AI |
| **Preconditions** | User has selected a lead or contact |
| **Postconditions** | Email is generated and ready to send |

**Main Flow:**

| Step | Action |
|---|---|
| 1 | User selects a lead/contact |
| 2 | User clicks “Generate Email” |
| 3 | User selects email type (intro, follow-up, proposal, etc.) |
| 4 | System retrieves lead context (company, role, interactions) |
| 5 | AI generates personalized email draft |
| 6 | System displays draft to user |
| 7 | User reviews and edits |
| 8 | User sends or saves draft |

# Entity Relationship Diagram (ERD)

```
+===================================================================================+
|                              ENTITY RELATIONSHIP DIAGRAM                           |
+===================================================================================+

+------------------+          +------------------+          +------------------+
|      USERS       |          |      LEADS       |          |    CONTACTS      |
+------------------+          +------------------+          +------------------+
| PK: id (uuid)    |---+      | PK: id (uuid)    |          | PK: id (uuid)    |
| email (unique)   |   |      | FK: user_id      |<---+     | FK: user_id      |<--+
| password_hash    |   |      | email            |    |     | FK: company_id   |   |
| full_name        |   |      | first_name       |    |     | email            |   |
| avatar_url       |   |      | last_name        |    |     | first_name       |   |
| role             |   |      | phone            |    |     | last_name        |   |
| timezone         |   |      | company_name     |    |     | phone            |   |
| email_verified   |   |      | job_title        |    |     | job_title        |   |
| created_at       |   |      | linkedin_url     |    |     | linkedin_url     |   |
| updated_at       |   |      | website          |    |     | address          |   |
+------------------+   |      | source           |    |     | custom_fields    |   |
         |             |      | score            |    |     | created_at       |   |
         |    +--------+      | status           |    |     | updated_at       |   |
         |    |               | enrichment_data  |    |     +------------------+   |
         |    |               | custom_fields    |    |               |            |
         |    |               | notes            |    |               |            |
         |    |               | created_at       |    |     +---------+------------+
         |    |               | updated_at       |    |     |
         |    |               +------------------+    |     |
         |    |                                       |     |
         |    +---------------------------------------+     |
         |    |                                             |
         |    |        +------------------+                 |
         |    |        |    COMPANIES     |                 |
         |    |        +------------------+                 |
         |    +------->| PK: id (uuid)    |-----------------+
         |             | FK: user_id      |<---+
         |             | name             |    |
         |             | domain (unique)  |    |
         |             | industry         |    |
         |             | size             |    |
         |             | linkedin_url     |    |
         |             | address          |    |
         |             | phone            |    |
         |             | enrichment_data  |    |
         |             | created_at       |    |
         |             | updated_at       |    |
         |             +------------------+    |
         |                      |              |
         |                      +--------------+
         |
+--------|----------------------------------------------------------------+
|        |                                                                |
|        v                                                                |
|  +------------------+          +------------------+                     |
|  |      DEALS       |          |   APPOINTMENTS   |                     |
|  +------------------+          +------------------+                     |
|  | PK: id (uuid)    |          | PK: id (uuid)    |                     |
|  | FK: user_id      |          | FK: user_id      |                     |
|  | FK: contact_id   |          | FK: event_type_id|                     |
|  | FK: company_id   |          | FK: contact_id   |                     |
|  | name             |          | title            |                     |
|  | amount           |          | description      |                     |
|  | currency         |          | start_time       |                     |
|  | stage            |          | end_time         |                     |
|  | probability      |          | timezone         |                     |
|  | expected_close   |          | status           |                     |
|  | description      |          | location         |                     |
|  | created_at       |          | meeting_url      |                     |
|  | updated_at       |          | google_event_id  |                     |
|  | closed_at        |          | outlook_event_id |                     |
|  +------------------+          | attendees (json) |                     |
|           |                    | created_at       |                     |
|           |                    | updated_at       |                     |
|           |                    +------------------+                     |
|           |                             ^                               |
|           |                             |                               |
|           |                    +------------------+                     |
```

```
|           |                    |   EVENT_TYPES    |                     |
|           |                    +------------------+                     |
|           |                    | PK: id (uuid)    |                     |
|           |                    | FK: user_id      |                     |
|           |                    | name             |                     |
|           |                    | slug (unique)    |                     |
|           |                    | description      |                     |
|           |                    | duration_minutes |                     |
|           |                    | color            |                     |
|           |                    | is_active        |                     |
|           |                    | availability_rules|                    |
|           |                    | buffer_times     |                     |
|           |                    | created_at       |                     |
|           |                    | updated_at       |                     |
|           |                    +------------------+                     |
+---------------------------------------------------------------------------+

+==================================================================================+
|                            SUPPORTING ENTITIES                                   |
+==================================================================================+

+------------------+     +------------------+     +------------------+
|      TASKS       |     |   ACTIVITIES     |     |      NOTES       |
+------------------+     +------------------+     +------------------+
| PK: id (uuid)    |     | PK: id (uuid)    |     | PK: id (uuid)    |
| FK: user_id      |     | FK: user_id      |     | FK: user_id      |
| FK: contact_id   |     | FK: contact_id   |     | FK: contact_id   |
| FK: deal_id      |     | FK: lead_id      |     | FK: lead_id      |
| title            |     | FK: deal_id      |     | FK: deal_id      |
| description      |     | type             |     | content          |
| priority         |     | title            |     | is_pinned        |
| status           |     | description      |     | created_at       |
| due_date         |     | metadata (json)  |     | updated_at       |
| completed_at     |     | created_at       |     +------------------+
| created_at       |     +------------------+
| updated_at       |
+------------------+

+------------------+     +------------------+     +------------------+
|  AVAILABILITY    |     |CALENDAR_CONNECTIONS|   |     AI_LOGS      |
+------------------+     +------------------+     +------------------+
| PK: id (uuid)    |     | PK: id (uuid)    |     | PK: id (uuid)    |
| FK: user_id      |     | FK: user_id      |     | FK: user_id      |
| day_of_week      |     | provider         |     | action_type      |
| start_time       |     | access_token     |     | model_used       |
| end_time         |     | refresh_token    |     | tokens_used      |
| is_available     |     | calendar_id      |     | input_data (json)|
| created_at       |     | token_expires_at |     | output_data(json)|
+------------------+     | is_active        |     | response_time_ms |
                         | created_at       |     | created_at       |
                         | updated_at       |     +------------------+
                         +------------------+

+------------------+     +------------------+     +------------------+
|      TAGS        |     |    LEAD_TAGS     |     |  CONTACT_TAGS    |
+------------------+     +------------------+     +------------------+
| PK: id (uuid)    |<----| FK: tag_id       |     | FK: contact_id   |
| FK: user_id      |     | FK: lead_id      |     | FK: tag_id       |---->
| name             |     +------------------+     +------------------+
| color            |
| entity_type      |
| created_at       |
+------------------+
```

## ERD Relationships Summary

| Relationship | Type | Description |
|---|---|---|
| Users → Leads | 1 : N | User owns many leads |
| Users → Contacts | 1 : N | User owns many contacts |
| Users → Companies | 1 : N | User owns many companies |
| Users → Deals | 1 : N | User owns many deals |
| Users → Appointments | 1 : N | User hosts many appointments |
| Users → Event_Types | 1 : N | User creates many event types |
| Users → Tasks | 1 : N | User creates many tasks |
| Companies → Contacts | 1 : N | Company employs many contacts |
| Companies → Deals | 1 : N | Company associated with many deals |
| Contacts → Deals | 1 : N | Contact involved in many deals |
| Contacts → Appointments | 1 : N | Contact attends many appointments |
| Event_Types → Appointments | 1 : N | Event type defines many appointments |
| Leads → Lead_Tags | 1 : N | Lead has many tags |
| Contacts → Contact_Tags | 1 : N | Contact has many tags |
| Tags → Lead_Tags | 1 : N | Tag applied to many leads |
| Tags → Contact_Tags | 1 : N | Tag applied to many contacts |

---

# Sequence Diagrams

## Sequence Diagram 1: Lead Capture and Enrichment

```
+--------+      +------------+      +----------+      +---------+      +-------------+      +----------+
|  User  |      |    Web     |      |   API    |      |   AI    |      | Enrichment  |      | Database |
|        |      | Interface  |      |  Server  |      | Module  |      |  Services   |      |          |
+---+----+      +-----+------+      +----+-----+      +----+----+      +------+------+      +----+-----+
    |                 |                  |                 |                  |                  |
    | 1. Enter LinkedIn URL              |                 |                  |                  |
    |---------------->|                  |                 |                  |                  |
    |                 |                  |                 |                  |                  |
    |                 | 2. POST /api/leads/parse           |                  |                  |
    |                 |----------------->|                 |                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 3. parseLinkedInProfile(url)       |                  |
    |                 |                  |---------------->|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  |                 | 4. Extract       |                  |
    |                 |                  |                 |    profile data  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 5. Parsed lead data                |                  |
    |                 |                  |<----------------|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 6. Initiate waterfall enrichment   |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 7. Apollo.io lookup                |                  |
    |                 |                  |---------------------------------->|                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 8. Company & contact data          |                  |
    |                 |                  |<----------------------------------|                  |
    |                 |                  |                 |                  |                  |
    |                 |                  |    [If Apollo data incomplete]     |                  |
    |                 |                  | 9. Clearbit lookup                 |                  |
    |                 |                  |---------------------------------->|                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 10. Additional data                |                  |
    |                 |                  |<----------------------------------|                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 11. scoreLead(enrichedData)        |                  |
    |                 |                  |---------------->|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 12. Score (0-100)                  |                  |
    |                 |                  |<----------------|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 13. categorizeLead(enrichedData)   |                  |
    |                 |                  |---------------->|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 14. Tags & category                |                  |
    |                 |                  |<----------------|                  |                  |
    |                 |                  |                 |                  |                  |
    |                 |                  | 15. Check duplicates               |                  |
    |                 |                  |--------------------------------------------------->|
    |                 |                  |                 |                  |                  |
    |                 |                  | 16. Duplicate status               |                  |
    |                 |                  |<---------------------------------------------------|
    |                 |                  |                 |                  |                  |
    |                 |                  | 17. INSERT lead                    |                  |
    |                 |                  |--------------------------------------------------->|
    |                 |                  |                 |                  |                  |
    |                 |                  | 18. Lead created                   |                  |
    |                 |                  |<---------------------------------------------------|
    |                 |                  |                 |                  |                  |
    |                 | 19. Lead response|                 |                  |                  |
    |                 |<-----------------|                 |                  |                  |
    |                 |                  |                 |                  |                  |
    | 20. Display enriched lead          |                 |                  |                  |
    |<----------------|                  |                 |                  |                  |
    |                 |                  |                 |                  |                  |
+---+----+      +-----+------+      +----+-----+      +----+----+      +------+------+      +----+-----+
|  User  |      |    Web     |      |   API    |      |   AI    |      | Enrichment  |      | Database |
+--------+      | Interface  |      |  Server  |      | Module  |      |  Services   |      +----------+
                +------------+      +----------+      +---------+      +-------------+
```

## Sequence Diagram 2: Appointment Booking

```
+----------+     +-------------+     +-----------+     +----------+     +----------+     +----------+
| Attendee |     | Booking     |     |   API     |     | Calendar |     | Database |     |  Notif.  |
|          |     | Page        |     |  Server   |     | Service  |     |          |     | Service  |
+----+-----+     +------+------+     +-----+-----+     +----+-----+     +----+-----+     +----+-----+
     |                  |                  |                |                |                |
     | 1. Access booking page             |                |                |                |
     |----------------->|                  |                |                |                |
     |                  |                  |                |                |                |
     |                  | 2. GET /api/event-types/{slug}   |                |                |
     |                  |----------------->|                |                |                |
     |                  |                  |                |                |                |
     |                  |                  | 3. Get event type details       |                |
     |                  |                  |-------------------------------->|                |
     |                  |                  |                |                |                |
     |                  |                  | 4. Event type data              |                |
     |                  |                  |<--------------------------------|                |
     |                  |                  |                |                |                |
     |                  | 5. Event type info               |                |                |
     |                  |<-----------------|                |                |                |
     |                  |                  |                |                |                |
     |                  | 6. GET /api/availability/slots   |                |                |
     |                  |----------------->|                |                |                |
     |                  |                  |                |                |                |
     |                  |                  | 7. Get user availability rules  |                |
     |                  |                  |-------------------------------->|                |
     |                  |                  |                |                |                |
     |                  |                  | 8. Availability config          |                |
     |                  |                  |<--------------------------------|                |
     |                  |                  |                |                |                |
     |                  |                  | 9. Fetch busy times             |                |
     |                  |                  |--------------->|                |                |
     |                  |                  |                |                |                |
     |                  |                  | 10. Busy slots |                |                |
     |                  |                  |<---------------|                |                |
     |                  |                  |                |                |                |
     |                  |                  | 11. Calculate available slots   |                |
     |                  |                  |                |                |                |
     |                  | 12. Available time slots         |                |                |
     |                  |<-----------------|                |                |                |
     |                  |                  |                |                |                |
     | 13. Display calendar               |                |                |                |
     |<-----------------|                  |                |                |                |
     |                  |                  |                |                |                |
     | 14. Select time slot               |                |                |                |
     |----------------->|                  |                |                |                |
     |                  |                  |                |                |                |
     | 15. Enter contact info             |                |                |                |
     |----------------->|                  |                |                |                |
     |                  |                  |                |                |                |
     |                  | 16. POST /api/appointments/book  |                |                |
     |                  |----------------->|                |                |                |
     |                  |                  |                |                |                |
     |                  |                  | 17. Verify slot available       |                |
     |                  |                  |-------------------------------->|                |
     |                  |                  |                |                |                |
     |                  |                  | 18. Confirmed  |                |                |
     |                  |                  |<--------------------------------|                |
     |                  |                  |                |                |                |
     |                  |                  | 19. INSERT appointment          |                |
     |                  |                  |-------------------------------->|                |
     |                  |                  |                |                |                |
     |                  |                  | 20. Appointment created         |                |
     |                  |                  |<--------------------------------|                |
     |                  |                  |                |                |                |
     |                  |                  | 21. Create Google/Outlook event |                |
     |                  |                  |--------------->|                |                |
     |                  |                  |                |                |                |
     |                  |                  | 22. Event IDs  |                |                |
     |                  |                  |<---------------|                |                |
     |                  |                  |                |                |                |
     |                  |                  | 23. Send confirmation (Host)    |                |
     |                  |                  |------------------------------------------------->|
     |                  |                  |                |                |                |
     |                  |                  | 24. Send confirmation (Attendee)|                |
     |                  |                  |------------------------------------------------->|
     |                  |                  |                |                |                |
```

```
     |                 | 25. Booking confirmation        |               |               |
     |                 |<-----------------|               |               |               |
     |                 |                  |               |               |               |
     | 26. Success page |                 |               |               |               |
     |<-----------------|                 |               |               |               |
     |                 |                  |               |               |               |
+----+-----+     +------+------+     +-----+-----+     +----+-----+     +----+-----+     +----+-----+
| Attendee |     | Booking     |     |   API     |     | Calendar |     | Database |     |  Notif.  |
+----------+     | Page        |     |  Server   |     | Service  |     +----------+     | Service  |
                 +-------------+     +-----------+     +----------+                      +----------+
```

## Sequence Diagram 3: AI Email Generation

```
+--------+      +------------+      +----------+      +---------+      +----------+
|  User  |      |    Web     |      |   API    |      |   AI    |      | Database |
|        |      | Interface  |      |  Server  |      | Module  |      |          |
+---+----+      +-----+------+      +----+-----+      +----+----+      +----+-----+
    |                 |                  |                 |                  |
    | 1. Select lead/contact             |                 |                  |
    |---------------->|                  |                 |                  |
    |                 |                  |                 |                  |
    | 2. Click "Generate Email"          |                 |                  |
    |---------------->|                  |                 |                  |
    |                 |                  |                 |                  |
    |                 | 3. GET /api/contacts/{id}          |                  |
    |                 |----------------->|                 |                  |
    |                 |                  |                 |                  |
    |                 |                  | 4. Fetch contact details           |
    |                 |                  |---------------------------------------->|
    |                 |                  |                 |                  |
    |                 |                  | 5. Contact data |                  |
    |                 |                  |<----------------------------------------|
    |                 |                  |                 |                  |
    |                 |                  | 6. Fetch interaction history       |
    |                 |                  |---------------------------------------->|
    |                 |                  |                 |                  |
    |                 |                  | 7. Activities & notes              |
    |                 |                  |<----------------------------------------|
    |                 |                  |                 |                  |
    |                 | 8. Contact context                 |                  |
    |                 |<-----------------|                 |                  |
    |                 |                  |                 |                  |
    | 9. Select email type               |                 |                  |
    |---------------->|                  |                 |                  |
    |                 |                  |                 |                  |
    |                 | 10. POST /api/ai/generate-email    |                  |
    |                 |----------------->|                 |                  |
    |                 |                  |                 |                  |
    |                 |                  | 11. Prepare prompt with context    |
    |                 |                  |---------------->|                  |
    |                 |                  |                 |                  |
    |                 |                  |                 | 12. Generate     |
    |                 |                  |                 |     personalized |
    |                 |                  |                 |     email        |
    |                 |                  |                 |                  |
    |                 |                  | 13. Email draft |                  |
    |                 |                  |<----------------|                  |
    |                 |                  |                 |                  |
    |                 |                  | 14. Log AI usage                   |
    |                 |                  |---------------------------------------->|
    |                 |                  |                 |                  |
    |                 |                  | 15. Logged      |                  |
    |                 |                  |<----------------------------------------|
    |                 |                  |                 |                  |
    |                 | 16. Email draft  |                 |                  |
    |                 |<-----------------|                 |                  |
    |                 |                  |                 |                  |
    | 17. Display draft                  |                 |                  |
    |<----------------|                  |                 |                  |
    |                 |                  |                 |                  |
    | 18. Edit and send                  |                 |                  |
    |---------------->|                  |                 |                  |
    |                 |                  |                 |                  |
    |                 | 19. POST /api/email/send           |                  |
    |                 |----------------->|                 |                  |
    |                 |                  |                 |                  |
    |                 |                  | 20. Log activity                   |
    |                 |                  |---------------------------------------->|
    |                 |                  |                 |                  |
    |                 | 21. Email sent   |                 |                  |
    |                 |<-----------------|                 |                  |
    |                 |                  |                 |                  |
    | 22. Confirmation                   |                 |                  |
    |<----------------|                  |                 |                  |
    |                 |                  |                 |                  |
+---+----+      +-----+------+      +----+-----+      +----+----+      +----+-----+
|  User  |      |    Web     |      |   API    |      |   AI    |      | Database |
```

```
+--------+       | Interface  |      |  Server  |      | Module  |      +----------+
                 +------------+      +----------+      +---------+
```

# Data Flow Diagrams (DFD)

## Level 0: Context Diagram

```
                              EXTERNAL ENTITIES
+-------------+    +-------------+    +-------------+    +-------------+
|    User     |    |  Attendee   |    |  Google     |    |  Outlook    |
| (Sales Rep) |    | (External)  |    |  Calendar   |    |  Calendar   |
+------+------+    +------+------+    +------+------+    +------+------+
       |                  |                  |                  |
       |                  |                  |                  |
Login  |  Lead Data       | Booking         |  Calendar        |  Calendar
Queries|  Updates         | Request         |  Sync            |  Sync
       |                  |                  |                  |
       v                  v                  v                  v
+=====================================================================+
||                                                                   ||
||                                                                   ||
||                   0. AI-POWERED CRM SYSTEM                        ||
||                                                                   ||
||                                                                   ||
+=====================================================================+
       |                  |                  |                  |
       |                  |                  |                  |
Dashboard               Confirmation      Events            Events
Reports                 Available Slots   Created           Created
AI Insights             Reminders
       |                  |                  |                  |
       v                  v                  v                  v
+------+------+    +------+------+    +------+------+    +------+------+
|    User     |    |  Attendee   |    |  Google     |    |  Outlook    |
+-------------+    +-------------+    |  Calendar   |    |  Calendar   |
                                      +-------------+    +-------------+

                   +-------------+    +-------------+    +-------------+
                   |  Apollo.io  |    |  Clearbit   |    |   Email     |
                   | (Enrichment)|    | (Enrichment)|    |  Service    |
                   +------+------+    +------+------+    +------+------+
                          |                  |                  |
                          |                  |                  |
                   Enrichment         Enrichment          Emails &
                   Request            Request             Reminders
                          |                  |                  |
                          v                  v                  v
                   +====================================+
                   ||  0. AI-POWERED CRM SYSTEM        ||
                   +====================================+
                          |                  |                  |
                          |                  |                  |
                   Company &          Enrichment         Delivery
                   Contact Data       Data               Status
                          |                  |                  |
                          v                  v                  v
                   +------+------+    +------+------+    +------+------+
                   |  Apollo.io  |    |  Clearbit   |    |   Email     |
                   +-------------+    +-------------+    |  Service    |
                                                         +-------------+
```

## Level 1: Main Processes

```
+==============================================================================+
|                            DFD LEVEL 1 - MAIN PROCESSES                       |
+==============================================================================+

EXTERNAL ENTITIES:
[User]  [Attendee]  [Google Calendar]  [Outlook]  [Enrichment APIs]  [Email]

                                    |
                                    v
+----------------------------------+----------------------------------+
|                                  |                                  |
|   +-----------+                  |                  +-----------+   |
|   |   1.0     |   Credentials    |   User Data     |   2.0     |   |
|   |  AUTH &   |<-----------------|---------------->|   LEAD    |   |
|   |  AUTHZ    |                  |                 | MANAGEMENT|   |
|   +-----------+                  |                 +-----------+   |
|        |                         |                      |          |
|        | Session Token           |                      | Lead     |
|        |                         |                      | Data     |
|        v                         |                      v          |
|   +----+----+                    |                 +----+----+     |
|   | D1:Users|                    |                 |D2: Leads|     |
|   +---------+                    |                 +---------+     |
|                                  |                      |          |
|                                  |                      |          |
|   +-----------+                  |                 +-----------+   |
|   |   3.0     |  Availability    |   Contact       |   4.0     |   |
|   |APPOINTMENT|<-----------------|---------------->| CONTACT & |   |
|   |SCHEDULING |  Booking Request |   Company Data  | COMPANY   |   |
|   +-----------+                  |                 | MGMT      |   |
|        |                         |                 +-----------+   |
|        | Appointments            |                      |          |
|        v                         |                      v          |
|   +----+--------+                |                 +----+-------+  |
|   |D5:Appointments|              |                 |D3: Contacts|  |
|   +-------------+                |                 |D4: Companies| |
|                                  |                 +------------+  |
|                                  |                                 |
|   +-----------+                  |                 +-----------+   |
|   |   5.0     |  Deal Updates    |   AI Request    |   6.0     |   |
|   |   DEAL    |<-----------------|---------------->|    AI     |   |
|   | MANAGEMENT|                  |                 | PROCESSING|   |
|   +-----------+                  |                 +-----------+   |
|        |                         |                      |          |
|        | Deals                   |                      | AI Logs  |
|        v                         |                      v          |
|   +----+----+                    |                 +----+----+     |
|   |D6: Deals|                    |                 |D8:AI Logs|    |
|   +---------+                    |                 +---------+     |
|                                  |                                 |
|   +-----------+                  |                                 |
|   |   7.0     |  Report Request  |                                 |
|   | REPORTING |<-----------------+                                 |
|   | ANALYTICS |                                                    |
|   +-----------+                                                    |
|        |                                                           |
|        | Activities                                                |
|        v                                                           |
|   +----+-------+                                                   |
|   |D7:Activities|                                                  |
|   +------------+                                                   |
|                                                                    |
+--------------------------------------------------------------------+

DATA STORES:
+----------+  +----------+  +----------+  +----------+
| D1:Users |  | D2:Leads |  |D3:Contacts|  |D4:Companies|
+----------+  +----------+  +----------+  +----------+

+-------------+  +----------+  +------------+  +----------+
|D5:Appointments|  | D6:Deals |  |D7:Activities|  |D8:AI Logs|
+-------------+  +----------+  +------------+  +----------+
```

## Level 2: Lead Management Process (2.0)

```
+==============================================================================+
|                    DFD LEVEL 2 - LEAD MANAGEMENT (Process 2.0)                |
+==============================================================================+

                        +----------+
                        |   User   |
                        +----+-----+
                             |
         +-------------------+-------------------+-------------------+
         |                   |                   |                   |
         | LinkedIn URL      | CSV File          | Export Request    |
         | Website/Manual    |                   |                   |
         v                   v                   v                   |
    +----+----+         +----+----+         +----+----+             |
    |   2.1   |         |   2.7   |         |   2.8   |             |
    | CAPTURE |         | IMPORT  |         | EXPORT  |             |
    |  LEAD   |         |   CSV   |         | LEADS   |             |
    +----+----+         +----+----+         +----+----+             |
         |                   |                   ^                   |
         | Raw Input         | Validated Rows    | Lead Data         |
         v                   v                   |                   |
    +----+----+              |              +----+----+              |
    |   2.2   |              |              |D2: Leads|<-------------+
    |  PARSE  |<-------------+              +---------+
    |  LEAD   |
    +----+----+
         |
         | Parse Request                          +-------------+
         +--------------------------------------->|  AI Module  |
         |                                        +------+------+
         | Parsed Data                                   |
         |<----------------------------------------------+
         v
    +----+----+
    |   2.3   |     Enrichment Request     +-------------+
    | ENRICH  |--------------------------->|  Apollo.io  |
    |  LEAD   |                            +------+------+
    +----+----+                                   |
         ^  |                                     |
         |  |     Company & Contact Data          |
         |  |<-----------------------------------+
         |  |
         |  |     Enrichment Request     +-------------+
         |  +--------------------------->|  Clearbit   |
         |                               +------+------+
         |                                      |
         |        Additional Data               |
         |<-------------------------------------+
         |
         v
    +----+----+
    |   2.4   |     Score Request          +-------------+
    |  SCORE  |--------------------------->|  AI Module  |
    |  LEAD   |                            +------+------+
    +----+----+                                   |
         ^                                        |
         |        Lead Score (0-100)              |
         |<---------------------------------------+
         |
         v
    +----+----+
    |   2.5   |     Check Existing
    | DETECT  |------------------------->+----------+
    |DUPLICATES|                         |D2: Leads |
    +----+----+                          +----+-----+
         ^                                    |
         |        Duplicate Status            |
         |<-----------------------------------+
         |
         v
    +----+----+     Categorize Request     +-------------+
    |   2.6   |--------------------------->|  AI Module  |
    |CATEGORIZE|                           +------+------+
    |  LEAD   |                                   |
    +----+----+                                   |
         ^                                        |
```

```
     |        Categories & Tags               |
     |<---------------------------------------+
     |
     v
+----+----+                          +----------+
|  SAVE   |------------------------->|D2: Leads |
|  LEAD   |                          +----------+
+---------+
     |
     |        Activity Log           +-------------+
     +------------------------------>|D7:Activities|
                                     +-------------+
```

## Level 2: Appointment Scheduling Process (3.0)

```
+============================================================================+
|             DFD LEVEL 2 - APPOINTMENT SCHEDULING (Process 3.0)              |
+============================================================================+

    +----------+                                          +----------+
    |   User   |                                          | Attendee |
    +----+-----+                                          +----+-----+
         |                                                     |
         | Availability Rules                                  | Date Range
         v                                                     | Request
    +----+----+                                                v
    |   3.1   |                                          +----+----+
    |CONFIGURE|                                          |   3.3   |
    |AVAILABIL|                                          |CALCULATE|
    +---------+                                          | SLOTS   |
         |                                               +----+----+
         | Availability Data                                  ^  |
         v                                                    |  |
    +----+-------+                                            |  |
    |D11:Availability|<---------------------------------------+  |
    +------------+       Availability Rules                      |
                                                                 |
    +----------+                                                 |
    |   User   |     Event Type Config                           |
    +----+-----+                                                 |
         |                                                       |
         v                                                       |
    +----+----+                                                  |
    |   3.2   |                                                  |
    | MANAGE  |                                                  |
    | EVENT   |                                                  |
    | TYPES   |                                                  |
    +----+----+                                                  |
         |                                                       |
         | Event Types                                           |
         v                                                       |
    +----+--------+                                              |
    |D10:EventTypes|<--------------------------------------------+
    +-------------+       Event Type Settings

                          +----+----+
                          |   3.5   |
                          |  SYNC   |     Fetch Busy Times
                          |CALENDARS|<-------------------+
                          +----+----+                    |
                               |                         |
         +---------------------------+                   |
         |                     |                         |
         v                     v                         |
    +----+------+          +-----+-----+                 |
    |  Google   |          |  Outlook  |                 |
    | Calendar  |          | Calendar  |                 |
    +----+------+          +-----+-----+                 |
         |                     |                         |
         | Busy Slots          | Busy Slots              |
         +------------+------------+                     |
                      |                                  |
                      v                                  |
             +----+----+                                 |
             |   3.3   |                                 |
             |CALCULATE|<--------------------------------+
             | SLOTS   |
             +----+----+
                  |
                  | Available Slots
                  v
             +----+-----+
             | Attendee |
             +----+-----+
                  |
                  | Booking Request
                  v
             +----+----+
             |   3.4   |
             | PROCESS |
             | BOOKING |
```

```
             +----+----+
                  |
     +------------+------------+------------+
     |            |            |            |
     v            v            v            v
+----+----+  +----+----+  +----+----+  +----+----+
|  SAVE   |  |  SYNC   |  |  SEND   |  |SCHEDULE |
|APPOINT- |  |CALENDARS|  | NOTIFS  |  |REMINDERS|
|  MENT   |  |   3.5   |  |   3.6   |  |   3.8   |
+----+----+  +----+----+  +----+----+  +----+----+
     |            |            |            |
     v            |            v            |
+----+---------+  |       +----+----+       |
|D5:Appointments| |       |  Email  |       |
+--------------+  |       | Service |       |
                  |       +---------+       |
                  v            |            |
             +----+-------+    v            |
             |  Google   |+----+----+       |
             | Calendar  ||  SMS    |       |
             +-----------+| Service |       |
             |  Outlook  |+---------+       |
             | Calendar  |                  |
             +-----------+                  |
                                            v
                                      +----+----+
                                      |  3.6    |
                                      |  SEND   |
                                      | NOTIFS  |
                                      +---------+
```

## Level 2: AI Processing (6.0)

```
+============================================================================+
|                      DFD LEVEL 2 - AI PROCESSING (Process 6.0)              |
+============================================================================+

                        +----------+
                        |   User   |
                        +----+-----+
                             |
    +------------------------+------------------------+------------------------+
    |                        |                        |                        |
    | Raw Text/URL           | Lead Data              | Email Request          |
    v                        v                        v                        |
+---+----+               +----+----+              +----+----+                  |
|  6.1   |               |   6.2   |              |   6.4   |                  |
|  LEAD  |               |  LEAD   |              |  EMAIL  |                  |
| PARSING|               | SCORING |              |GENERATION                  |
+---+----+               +----+----+              +----+----+                  |
    |                        |                        |                        |
    | Prompt                 | Scoring Prompt         | Email Prompt           |
    v                        v                        v                        |
+---+--------+           +----+--------+          +----+--------+              |
|   OpenAI   |           |   OpenAI   |          |   OpenAI   |              |
|   GPT-4    |           |   GPT-4    |          |   GPT-4    |              |
+---+--------+           +----+--------+          +----+--------+              |
    |                        |                        |                        |
    | Extracted Data         | Score                  | Email Draft            |
    v                        v                        v                        |
+---+----+               +----+----+              +----+----+                  |
| Parsed |               |  Lead   |              |Generated|                  |
|  Lead  |               |  Score  |              |  Email  |                  |
+---+----+               +----+----+              +----+----+                  |
    |                        |                        |                        |
    |    +-------------------+------------------------+                        |
    |    |                                                                     |
    |    v                                                                     |
    | +--+---+                                                                 |
    | | 6.6  |                                                                 |
    | |TOKEN |                                                                 |
    | |MGMT  |                                                                 |
    | +--+---+                                                                 |
    |    |                                                                     |
    |    | Usage Tracking                                                      |
    |    v                                                                     |
    | +--+-------+                                                             |
    | |D8:AI Logs|                                                             |
    | +----------+                                                             |
    |                                                                          |
    +------------------------+------------------------+------------------------+
                             |                        |
                             | Lead for Tagging       | Scheduling Query
                             v                        v
                        +----+----+              +----+----+
                        |   6.3   |              |   6.5   |
                        |  LEAD   |              |  SMART  |
                        |CATEGORY |              |SCHEDULE |
                        +----+----+              +----+----+
                             |                        |
                             | Categorization         | Scheduling
                             | Prompt                 | Prompt
                             v                        v
                        +----+--------+          +----+--------+
                        |   OpenAI   |          |   OpenAI   |
                        |   GPT-4    |          |   GPT-4    |
                        +----+--------+          +----+--------+
                             |                        |
                             | Tags & Category        | Time Suggestions
                             v                        v
                        +----+----+              +----+----+
                        |Categorized              |Suggested|
                        |  Lead   |              | Times   |
                        +---------+              +---------+
```

## Technology Stack Summary

| Layer | Technology |
|---|---|
| Frontend | Next.js 14+, React 18+, TypeScript, Tailwind, Shadcn |
| Backend | Next.js API Routes, Drizzle ORM |
| Database | PostgreSQL (Supabase) |
| Authentication | Supabase Auth |
| AI | OpenAI GPT-4o |
| Calendar | Google Calendar API, Microsoft Outlook API |
| Enrichment | Apollo.io, Clearbit |
| Notifications | Resend/SendGrid (Email), Twilio (SMS) |
| Deployment | Vercel |

---

## References


- Next.js Documentation: https://nextjs.org/docs
- Supabase Documentation: https://supabase.com/docs
- OpenAI API Reference: https://platform.openai.com/docs
- Drizzle ORM Documentation: https://orm.drizzle.team/docs
- Google Calendar API: https://developers.google.com/calendar
- Microsoft Graph API: https://docs.microsoft.com/en-us/graph


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